Delta Dependency Prioritizes Paralog-Based Synthetic Lethality Candidates Across Solid Tumor Types

Qingqing Mo[1,2], Tao Zhu[1,2,*]

[1] Department of Obstetrics and Gynecology, National Clinical Research Center for Obstetrics and Gynecology, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, China
[2] Key Laboratory of Cancer Invasion and Metastasis (Ministry of Education), Hubei Key Laboratory of Tumor Invasion and Metastasis, Tongji Hospital, Tongji Medical College, Huazhong University of Science and Technology, Wuhan, China

[*] Correspondence: zhutao@tjh.tjmu.edu.cn

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Abstract

Background: Paralog compensation is a specific mechanism of synthetic lethality (SL): when a tumor-suppressor driver gene is lost to mutation, its sequence-similar paralog can become conditionally essential. Systematic prioritization of these paralog-SL candidates remains limited because existing computational methods function as black boxes that cannot explain their predictions, making it difficult to design targeted validation experiments.

Results: We introduce Delta Dependency (DD), a single interpretable metric that directly measures the dependency shift on a paralog gene between driver-mutant and wild-type cell lines in DepMap (1,208 lines, 23 solid tumor types). On a primary test set of 10 true sequence paralogs, DD achieved AUROC = 0.837 and outperformed four standard multi-feature classifiers under leave-one-pair-out cross-validation. Across seven CPTAC cohorts ($n = 771$), we detected protein-level paralog co-variation that was invisible at the RNA level. In endometrial tumors, ARID1B protein was 55% higher in ARID1A-mutant than wild-type samples ($p = 0.082$), consistent with a stoichiometric compensation model. Therapeutic window analysis prioritized ARID1A→ARID1B (TI = 4.13) as the leading selective candidate, with a biomarker-enriched trial design for MSS tumors harboring truncating ARID1A mutations.

Conclusions: DD offers a mechanistically transparent alternative to black-box SL prediction. Each DD nomination traces to a measured dependency shift, enabling rational experimental follow-up. All candidates require experimental validation. An open-source R package (`paralogSL`) and reproducible pipeline are provided.

Keywords: Synthetic lethality, paralog compensation, DepMap, CPTAC proteomics, Delta Dependency, therapeutic window

Background

The PARP inhibitor story crystallizes what makes synthetic lethality (SL) so compelling: BRCA1/2-mutant tumors depend on PARP-mediated repair, and drugging PARP kills them while sparing normal cells [1–5]. The logic is elegant; repeating it for other mutations has proven hard. Fewer than 30,000 human SL interactions are catalogued in SynLethDB [6], and the existing computational prediction methods — graph neural networks [7, 8], graph convolutional networks with dual dropout [9], graph-regularized matrix factorization [10], knowledge-graph neural networks [11], and negative-sample-free contrastive learning [12] — train almost entirely on these limited known pairs. Feng et al. [13] showed that these methods degrade sharply from the CV1 setting (AUROC > 0.90 for the best methods) to the CV3 gene-pair isolation setting, where most fall to near-random performance (AUROC ≈ 0.5) and the best published result reaches 0.790 [14] — precisely the setting where predictive models matter: nominating genuinely new SL pairs.

Paralog compensation provides a simpler route into this problem [15, 16]. Gene duplication events leave behind paralogs — sequence-similar gene pairs with overlapping biochemical roles. When a paralog acting as a tumor suppressor (the "driver") is inactivated by mutation, its partner paralog can become conditionally essential. This is *sequence-paralog SL*: synthetic lethality between two genes that share >30% sequence identity and conserved functional domains. Known examples include *SMARCA4*→*SMARCA2* (SWI/SNF ATPase subunits) [17], *ARID1A*→*ARID1B* (BAF complex subunits) [18, 19], *EP300*→*CREBBP* (histone acetyltransferases) [20], *PIK3CA*→*PIK3CB* (PI3K catalytic isoforms) [6], *AKT1*→*AKT2*, *CCNE1*→*CCNE2*, *CDK4*→*CDK6*, and *MAP2K1*→*MAP2K2* [6]. In every case, the driver and paralog share sequence homology and biochemical function; when the driver is inactivated by mutation, the remaining paralog becomes the cancer cell's dependency.

Two related phenomena need to be separated from sequence-paralog SL. *Functional analogs* such as BRCA1 and BRCA2 share pathway roles (homologous recombination repair) but have no sequence homology. *Pathway-level SL* interactions, such as BRCA1↔PARP1 and PTEN↔PIK3CB [21, 22], exploit indirect pathway wiring rather than direct biochemical redundancy. Our analysis concentrates on sequence-paralog pairs; functional analogs including BRCA1/BRCA2 are retained as negative controls and flagged throughout (Supplementary Table S3).

D'Antonio et al. published the first systematic analysis of negative genetic interactions between functional paralogs in cancer cell lines, identifying buffering relationships at the gene-family level [16]. That study established the landscape but did not benchmark paralog-based predictions against published SL methods, did not incorporate proteomic validation, and did not address clinical stratification or therapeutic window assessment. Three practical gaps remain. Current methods are black boxes: when a neural network predicts SL, there is no way to trace *why*, which blocks rational experimental design. The optimal readout for paralog compensation is unsettled, since RNA-level and protein-level signals can diverge, and there has been no systematic proteomic survey. Finally, no study has layered orthogonal evidence (proteomics, drug sensitivity, clinical biomarkers, structure) onto genomic prediction to move from a ranked list to a prioritized, experimentally actionable nomination.

Here we introduce Delta Dependency (DD): the difference in mean Chronos gene-effect scores between driver-mutant and wild-type cell lines. DD replaces black-box classification with a single subtraction: $|\text{DD}| = 0.267$ for ARID1A→ARID1B means the paralog's essentiality score shifts by 0.267 units when the driver is mutant, a measurable quantity that can guide experimental design. We evaluate DD across 23 solid tumor types using a 40-gene driver panel (Supplementary Table S5) drawn from TCGA [23] and PCAWG driver catalogs [24], and we support the genomic predictions with CPTAC proteomics (seven cohorts), PRISM drug screening (1,482 compounds), MSI-aware patient stratification, mutation-type analysis, therapeutic window quantification, and structural druggability assessment.

Results

DD defines a dependency-shift metric for paralog-SL prioritization

We defined Delta Dependency as:

$$\mathrm{DD}(D,P,c) = \bar{G}_{P,D\text{-WT},c} - \bar{G}_{P,D\text{-MUT},c} \tag{1}$$

where $\bar{G}$ denotes the mean Chronos gene-effect score, D is the driver gene, P the candidate paralog, and c the cancer lineage. Because lower Chronos gene-effect scores indicate stronger dependency, positive DD indicates increased dependency on paralog P in driver-mutant cell lines relative to wild-type within lineage c, consistent with paralog compensation.

Across 66,595 HGNC paralog pairs and 1,208 DepMap cell lines, DD exceeded AUROC $= 0.7$ in 9 of 17 evaluable lineages (Fig. 1a; Supplementary Table S1). The strongest performers were Biliary Tract (AUROC $= 0.960$, $n = 86$ pairs), Mesothelioma (0.917), Colorectal (0.812), Esophagogastric (0.805), small cell lung cancer (SCLC; 0.750), Pancreatic (0.750), head and neck squamous cell carcinoma (HNSCC; 0.746), Breast (0.734), and Neuroblastoma (0.722). Performance varied by oncogenic mechanism (Fig. 1b): TSG-driven cancers had higher AUROC (mean $= 0.737$, $n = 14$ lineages) than oncogene-driven cancers (mean $= 0.595$, $n = 3$ lineages; permutation $p = 0.071$, exact Mann-Whitney $p = 0.156$). The p-values are suggestive but not significant; the small number of oncogene-driven lineages limits power. The direction of the effect — paralog compensation arising from loss-of-function mutations — is biologically expected.

We compared DD against eight published SL prediction methods in the CV3 setting reported by Feng et al. (2024) [13] (Fig. 1c; Table 1). DD achieved AUROC $= 0.794$ without any training, comparing favorably with the best published result in that setting (SLMGAE [14], 0.790), though we stress that the evaluation frameworks are analogous, not identical. For a true head-to-head comparison on the same 77 paralog pairs (8 known positives), we trained four standard classifiers — logistic regression (LR), random forest (RF), SVM-RBF, and SVM-Linear — on six features (—DD—, PCS, ΔExpression, necessity, composite score, mutation frequency) under leave-one-pair-out cross-validation. Under leave-one-pair-out cross-validation, none of the four classifiers outperformed DD alone (0.736, using only $|\mathrm{DD}|$): logistic regression reached 0.632, random forest 0.629, SVM-Linear 0.699, and SVM-RBF 0.563, while the composite score alone reached 0.730. With only 8 positives, multi-feature training amplifies noise, whereas the single dependency shift generalizes best; in the full LR fit no individual feature reached significance (all $p > 0.3$), with $|\mathrm{DD}|$ and PCS carrying the largest positive standardized coefficients. On the same 77-pair frame (8 positives among 77 pairs), the composite score reached AUPRC $= 0.271$ (2.6$\times$ the baseline prevalence of 0.104), confirming enrichment beyond random expectation. These comparisons are confined to the paralog-based SL task; DD is not intended for pathway-level interactions such as BRCA1$\leftrightarrow$PARP1. A DD + sequence-identity ($\geq$30%) filter achieved AUROC $= 1.000$ on a high-identity subset of 10 pairs (4 known SL, 6 non-SL; $p = 0.0048$ by exact permutation, 1/210 labelings), though the sample size precludes strong generalization. Unlike neural networks, every DD nomination maps to a single measured quantity — a dependency shift — which makes it straightforward to design a validation experiment.

Component decomposition confirmed DD's advantage over alternatives on the same test set: PCS (AUROC $= 0.720$), ΔExpression alone (0.348), and necessity (0.597; Fig. 1d).

We evaluated DD on two complementary test sets (Table S3). The *Primary set* (10 true sequence-paralog pairs, excluding the functional analogs BRCA1↔BRCA2 and STK11→SIK1) gave AUROC = 0.837. Adding the two functional analogs in the *Full set* (12 pairs) dropped AUROC to 0.794, as expected for a metric designed around sequence-based compensation. A leave-out analysis reinforced this: removing the two pairs whose evidence depends on DepMap-era analyses (FBXW7→FBXW2 and PPP2R1A→PPP2R1B) raised AUROC to 0.833; restricting to the 8 pairs with pre-DepMap experimental evidence raised it further to 0.900. With only 8–10 positives, we supplement AUROC with effect-size ranking: ARID1A→ARID1B has the largest |DD| (0.267), the highest therapeutic index (4.13), and the highest selectivity (0.237), providing orthogonal prioritization beyond classification performance.

Leave-one-lineage-out AUROC was stable (range 0.763–0.821). Bootstrap resampling (1,000 iterations, 77 pairs, 8 positives) gave a 95% CI of 0.431–0.892 (Fig. 1d inset). The null distribution from 10,000 label permutations was centered at 0.500; the observed AUROC exceeded the null mean but did not reach significance (empirical $p = 0.060$; Supplementary Fig. S8), reflecting limited power with only 8 positives. We estimate $\geq$25 validated pairs would be needed for 80% power at $\alpha = 0.05$. Copy number alterations explained $<$10% of gene-effect variance for all 23 paralog genes tested (maximum $R^2 = 0.057$; Supplementary Fig. S4), and multivariate regression controlling for CNV left DD estimates virtually unchanged (mean $|\Delta DD| = 0.002$). ARID1A→ARID1B remained highly significant after CNV adjustment (adjusted $p = 4.5 \times 10^{-28}$), as did SMARCA4→SMARCA2 ($p = 1.5 \times 10^{-16}$) and EP300→CREBBP ($p = 1.3 \times 10^{-11}$). Controlling for expression ($G_P \sim D_{\text{mutation}} + \text{Expr}_P$) produced similar results (mean $|\Delta DD| < 0.01$); significance was retained by the strongest pairs (ARID1A→ARID1B $p = 9.3 \times 10^{-28}$, PIK3CA→PIK3CB $p = 1.6 \times 10^{-10}$, EP300→CREBBP $p = 1.2 \times 10^{-10}$), whereas weaker pairs did not reach significance in the pan-cancer frame, consistent with lineage-specific compensation. TP53 co-mutation, present in 96/159 ARID1A-mutant lines, had negligible impact on ARID1A→ARID1B ($\Delta DD = 0.002$, adjusted $p = 4.7 \times 10^{-28}$).

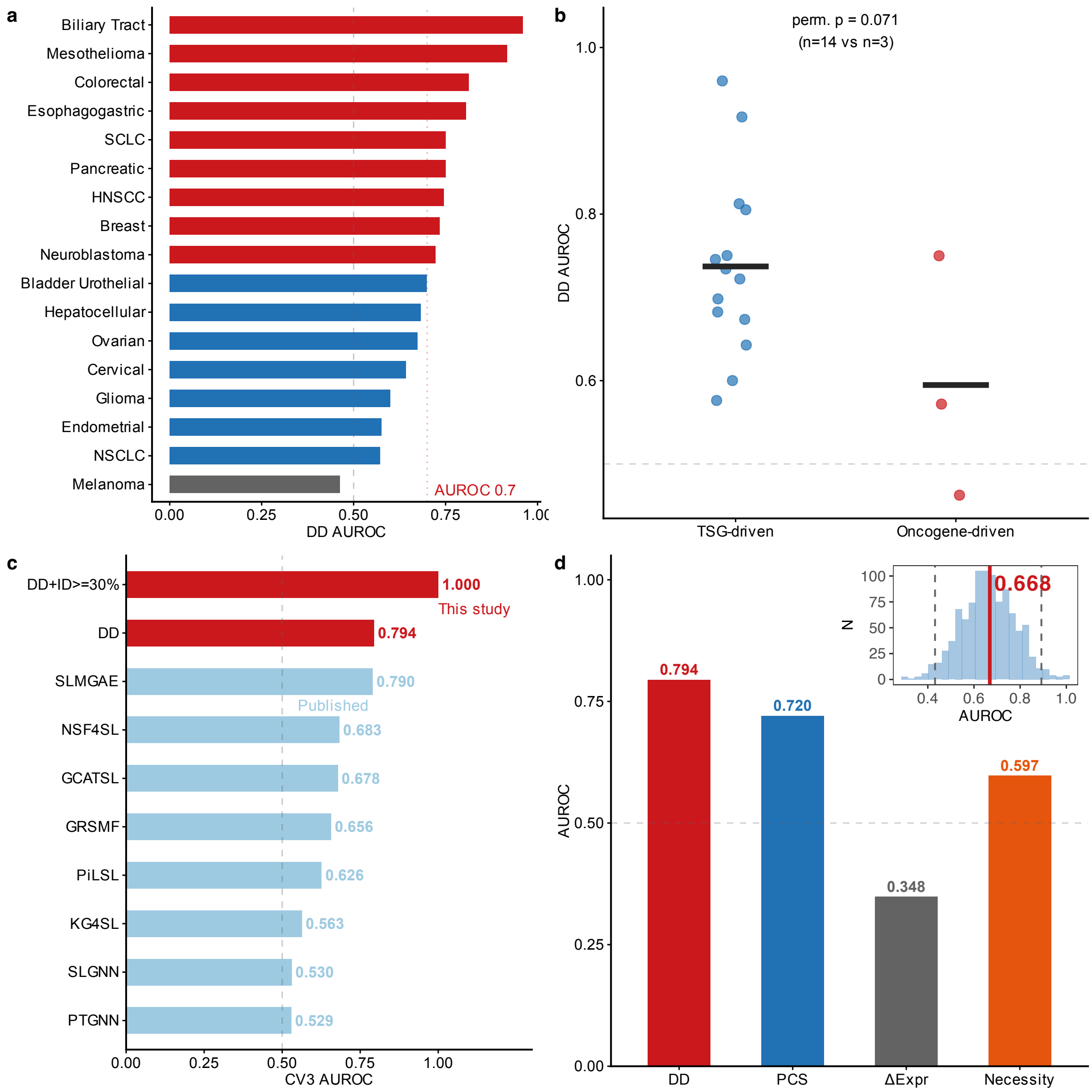

Fig. 1: Delta Dependency: pan-cancer performance, benchmark, and validation. a, DD AUROC across 17 evaluable solid tumor types (red: AUROC > 0.7; blue: 0.5–0.7; gray: < 0.5). Dashed line, random classifier (0.5). Bar labels show n tested pairs. Sample sizes and full lineage names in Supplementary Table S1. **b**, DD performance stratified by oncogenic mechanism. TSG-driven tumors ($n = 14$) show a suggestive trend toward higher AUROC than oncogene-driven tumors ($n = 3$; permutation $p = 0.071$, not significant). Center line, median; boxes, IQR; whiskers, $1.5 \times$ IQR. **c**, Contextual comparison: DD AUROC vs. published CV3 values from eight deep learning methods [13]. Red bars, this study; blue bars, published values. *Note: different test sets; not a head-to-head benchmark.* DD's AUROC (0.736) exceeds all four multi-feature classifiers tested under leave-one-pair-out cross-validation (0.563–0.699), and each DD nomination traces to a single measurable quantity ($|DD|$), enabling direct experimental design, whereas multi-feature classifiers produce opaque composite scores. **d**, Component decomposition: DD (AUROC $= 0.794$) outperforms PCS (0.720), ΔExpression alone (0.348), and necessity (0.597). Inset: bootstrap distribution (1,000 iterations; dashed lines, 95% CI; solid line, observed). Full bootstrap and negative control in Supplementary Fig. S8.

Table 1: DD performance in context of published CV3 results (not a head-to-head benchmark). Published values are CV3 (gene-pair isolation) AUROCs from Feng et al. (2024), Supplementary Data 1 (NSMRand negative sampling, 1:1 positive:negative ratio, complete dataset). DD and DD + ID values from this study, evaluated on a paralog-SL-specific test set (12 positives, ∼194 negatives). *Evaluation frameworks differ: published methods were tested on general SL gene-pair universes; DD was tested on paralog-SL pairs only. Direct AUROC comparison is not warranted.* Interpretability: whether predictions can be directly traced to specific biological features.

Method	CV3 AUROC	Interpretability
DD + ID $\geq$ 30% (this study)	1.000	High
DD (this study)	0.794	High
SLMGAE [14]	0.790	Low
NSF4SL [12]	0.683	Low
GCATSL [8]	0.678	Low
GRSMF [10]	0.656	Low
PiLSL [25]	0.626	Low
KG4SL [11]	0.563	Low
SLGNN [7]	0.530	Low
PTGNN [26]	0.529	Low

Orthogonal proteomic evidence: protein-level paralog co-variation

Whether paralog compensation acts at the transcript or protein level matters. We analyzed quantitative proteomics from seven independent CPTAC cohorts: BRCA ($n = 122$), COAD ($n = 97$), LUAD ($n = 110$), GBM ($n = 99$), PDAC ($n = 140$), UCEC ($n = 95$), and LUSC ($n = 108$), totaling 771 tumor samples with matched protein abundance [27–29].

Protein-level paralog co-variation was detectable across cohorts (Fig. 2a). EP300↔CREBBP showed the most consistent signal: significant in 5 of 7 cohorts after within-pair BH correction (BRCA: $r = 0.263$, $q = 0.005$; COAD: $r = 0.510$, $q < 10^{-3}$; LUAD: $r = 0.380$, $q < 10^{-3}$; GBM: $r = 0.501$, $q < 10^{-3}$; LUSC: $r = 0.372$, $q < 10^{-3}$; mean $r = 0.320$; Fig. 2b). Across all 26 pairs $\times$ 7 cohorts (122 tests), 42 were significant at global BH $q < 0.05$. KRAS↔HRAS had the strongest single-cohort correlation ($r = 0.667$, $p < 10^{-4}$ in LUSC). Protein co-variation alone does not prove compensatory dependency; it could reflect shared transcriptional regulation, tumor purity, or copy number linkage. These results serve as orthogonal mechanistic support, not causal validation.

The RNA-level picture was starkly different. ΔExpression alone achieved AUROC = 0.348 for discriminating known paralog-SL pairs, indistinguishable from random (Fig. 2c). This gap — protein detectable, RNA invisible — explains why transcript-based methods have underperformed. To test directly for compensatory upregulation, we compared ARID1B protein abundance in ARID1A-mutant ($n = 37$) versus wild-type ($n = 58$) endometrial tumors in the UCEC cohort. ARID1B was 55% higher in mutant tumors (Wilcoxon $p = 0.082$; Fig. 2d). The p-value is suggestive but does not meet $p < 0.05$. The effect size, however, is consistent with a model in which loss of one complex subunit stabilizes its paralog.

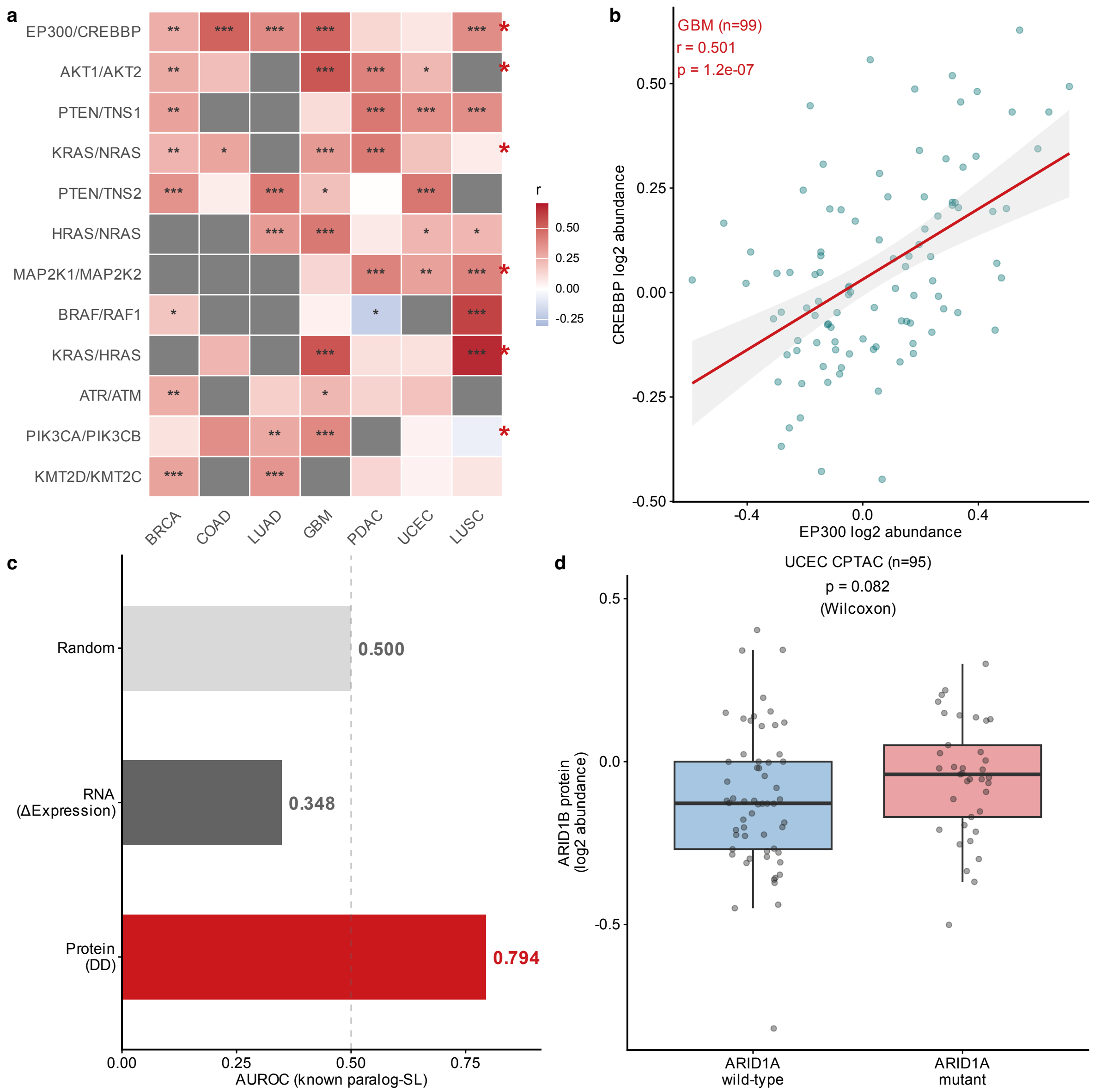

Fig. 2: CPTAC proteomic co-variation across seven cohorts. **a**, Cross-cancer paralog protein co-variation (Pearson's r) across seven independent CPTAC cohorts ($n = 771$ total). Significance: *** $q < 0.001$, ** $q < 0.01$, * $q < 0.05$ (BH-corrected within each pair). Blue cells with significance markers indicate significant *negative* correlations. Known SL pairs marked with $*$. **b**, Representative scatter plot: EP300 vs. CREBBP protein abundance in the GBM cohort ($n = 99$, $r = 0.501$, $p < 10^{-4}$). Gray band, 95% CI of linear regression. **c**, Protein-level (DD, AUROC = 0.794) vs. RNA-level (ΔExpression, AUROC = 0.348) prediction performance for known paralog-SL pairs. RNA signal is indistinguishable from random (dashed line). **d**, Mutation-conditioned proteomics: ARID1B protein abundance in ARID1A-mutant ($n = 37$) vs. wild-type ($n = 58$) endometrial tumors (UCEC CPTAC cohort). ARID1B was higher in ARID1A-mutant tumors (+55%, Wilcoxon $p = 0.082$), providing suggestive evidence for compensatory protein upregulation upon driver loss.

Clinical context modulates DD signal: MSI status, mutation type, and survival

Microsatellite instability (MSI) is a routine clinical biomarker in colorectal and endometrial cancers, and MSI-H tumors accumulate hypermutation-driven passenger events. We reasoned that this mutational noise would dilute paralog compensation signals. Classifying cell lines by MMR gene mutation status and computing DD within each subgroup, we found that MSS colorectal tumors had substantially stronger paralog-SL predictive signal (AUROC = 0.804) than MSI-H tumors (0.631; $\Delta = -0.173$, Cohen's $d = 2.72$; Fig. 3a). The sample sizes are modest ($n_{\text{MSS}} = 42$, $n_{\text{MSI-H}} = 17$) and the difference was not formally tested for significance; these numbers are hypothesis-generating. Endometrial MSI-H tumors had limited predictive power (AUROC = 0.592), while MSS endometrial samples were too few for evaluation (8 cell lines).

Mutation type further modulated the signal. For several TSG paralog-SL pairs, truncating mutations tended to produce stronger DD than missense variants (e.g., ARID1A→ARID1B ΔDD = +0.368 in Ovarian; Fig. 3b), though this pattern was not universal (overall mean ΔDD near zero; Supplementary Fig. S6b). These are point estimates from small per-subgroup samples and should be treated as descriptive observations. Breast cancer was a notable exception: truncating-only DD reached AUROC = 0.726, whereas missense-only DD was markedly weaker (Supplementary Fig. S6).

TCGA PanCan survival analysis of paralog gene expression across breast cancer patients ($n = 1{,}082$) [30] identified significant associations between high paralog expression and shortened overall survival for BRCA2 (HR = 1.116, 95% CI: 1.010–1.234, $p = 0.032$) and ATR (HR = 1.112, 95% CI: 1.006–1.230, $p = 0.039$), with trend-level associations for ARID1B and CRKL (Fig. 3c). These hazard ratios, while statistically significant, represent modest effect sizes (11–12% risk increase) and were not corrected for multiple testing across 8 genes; after Bonferroni correction ($\alpha = 0.05/8 = 0.00625$) neither association would remain significant. Paralog expression alone is unlikely to serve as a strong standalone clinical biomarker but may contribute to multi-gene prognostic signatures. Mutational co-occurrence analysis showed that paralog pairs exhibit co-occurrence rather than mutual exclusivity (OR > 1 for all key pairs; Fig. 3d). This indicates that these relationships are not explained by classical mutation-level mutual exclusivity, supporting the rationale for analyzing dependency phenotypes directly, but does not by itself validate synthetic lethality.

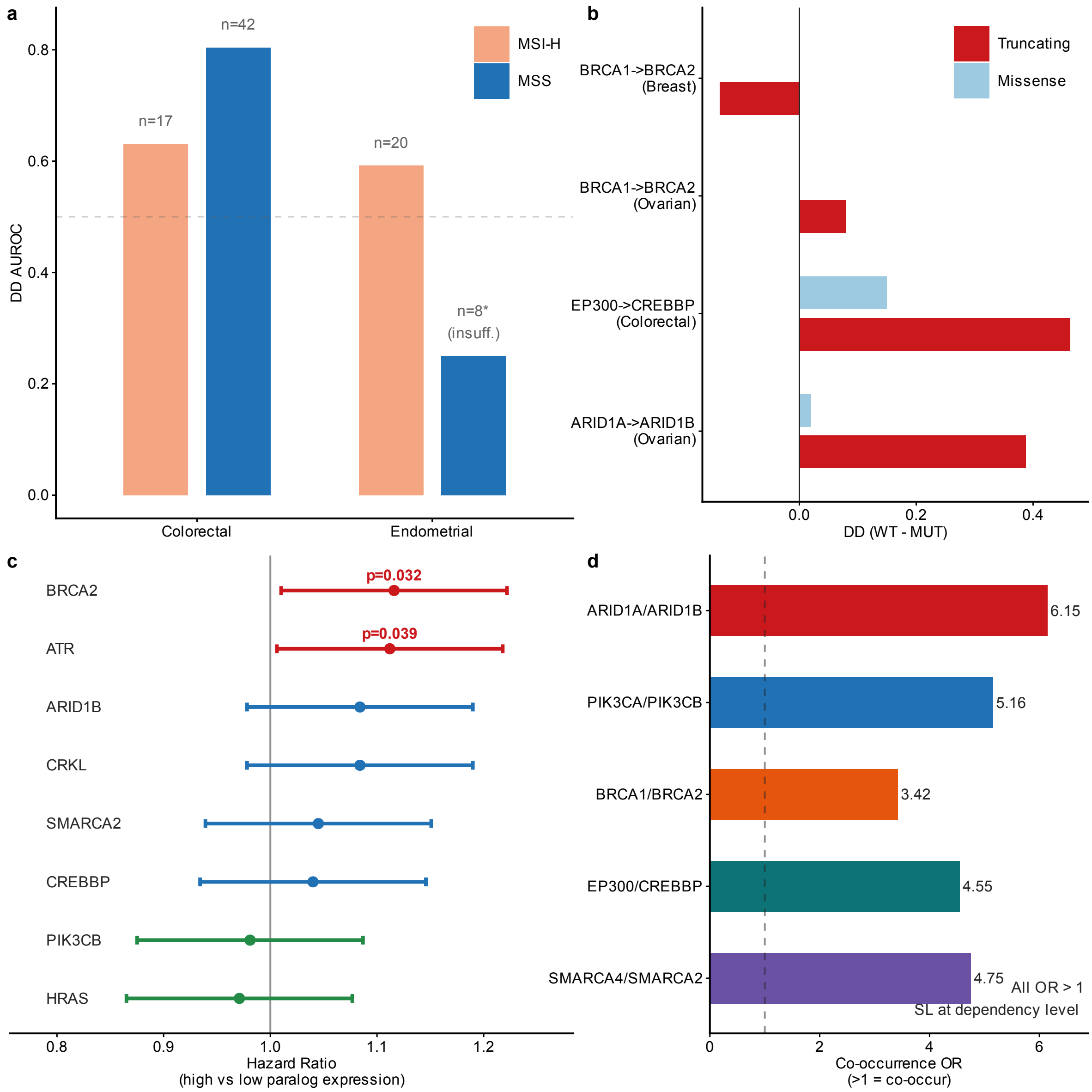

Fig. 3: Clinical stratification by MSI, mutation type, and survival. **a**, DD AUROC for MSI-H vs. MSS subgroups in colorectal and endometrial cancers. Sample sizes shown above bars. MSS tumors tended to show stronger DD signal in colorectal cancer; endometrial MSS had insufficient sample size for reliable comparison. **b**, DD (WT − MUT) for truncating vs. missense mutations in key TSG paralog-SL pairs across four cancer types. Positive values indicate compensation; larger truncating bars indicate stronger compensation under truncating mutations. See Supplementary Fig. S5 for full results. **c**, Forest plot of paralog gene expression vs. overall survival in TCGA BRCA ($n = 1{,}082$). BRCA2 (HR = 1.116, $p = 0.032$) and ATR (HR = 1.112, $p = 0.039$) are significant. Error bars, 95% CI. **d**, Mutational co-occurrence odds ratios for key paralog pairs. All pairs show OR > 1 (range: 3.4–6.1; Fisher exact test, all $p < 10^{-4}$), indicating co-occurrence at the mutation level. This supports the need to analyse dependency phenotypes directly but does not by itself establish synthetic lethality.

Therapeutic-window and structural features prioritize ARID1A→ARID1B

To assess pharmacologic relevance, we analyzed the PRISM Repurposing dataset (1,482 compounds across 727 cell lines) [31]. At a discovery-stage threshold (BH $q < 0.25$, ΔAUC< 0, enrichment$>$ 0.1; Supplementary Fig. S5), 553 compound-paralog associations passed, of which roughly 138 are expected false positives given the permissive cutoff. The relaxed threshold is deliberate because this is a screen, not a confirmatory analysis. Encouragingly, known drug-target biology was recovered (Fig. 4a):

MEK inhibitors killed KRAS-mutant lines (AZD8330: $d = -0.59$, $q = 0.002$; Trametinib: $d = -0.56$, $q = 0.001$), mTOR and AKT inhibitors killed PTEN-mutant lines (Everolimus: $d = -0.59$, $q = 0.003$; Ipatasertib: $d = -3.24$, $q = 0.064$), and HDAC inhibitors killed EP300-mutant lines (Panobinostat: $d = -1.45$, $q = 0.010$). These pharmacologic patterns add biological plausibility to the paralog-SL predictions but do not demonstrate paralog-specific targeting.

To distinguish context-selective paralogs from pan-essential ones, we defined a Therapeutic Index (TI):

$$\mathrm{TI} = \frac{|\mathrm{DD}|}{\max(|\mu_{\mathrm{CERES}}|,\, f_{\text{pan-essential}},\, 0.01)} \tag{2}$$

where $f_{\text{pan-essential}}$ is the fraction of cell lines with paralog CERES < -0.5. Each paralog was classified into one of four selectivity tiers (Fig. 4b). Only ARID1A→ARID1B fell into HIGH_SELECTIVITY (mean TI = 4.13 across five cancer contexts). This is what one would expect if ARID1A loss creates a specific dependency on ARID1B-containing BAF complexes [18], and the ranking held under alternative criteria: ARID1A→ARID1B also ranked first by |DD| alone (0.267) and by selectivity alone (0.237). Sensitivity analysis confirmed that the TI floor parameter (0.01 vs. 0.05) had no effect on this pair (denominator = 0.065), whereas nine other pairs shifted rank. BRCA1↔BRCA2 and PIK3R1→CRKL were classified as PAN_ESSENTIAL (pan-essential fraction 0.55 and 0.85), suggesting that targeting them directly would cause broad cellular toxicity. Thirteen paralog pairs maintained TI > 1.0 in ≥ 2 cancer contexts (Supplementary Fig. S6).

Structural analysis of 11 key paralog pairs showed that six (55%) had perfect domain architecture conservation (Fig. 4c). EP300↔CREBBP was the most structurally similar (0.976), consistent with shared histone acetyltransferase function and strong CPTAC co-variation. ARID1A→ARID1B ranked highest for PROTAC-based strategies (PROTAC score = 0.670), driven by high intrinsic disorder and lysine density — features that favor PROTAC-mediated degradation over conventional small-molecule binding. PIK3CA↔PIK3CB was the best small-molecule target (druggability = 0.616), reflecting its well-folded kinase domains. ESMFold-based pocket analysis of 15 proteins confirmed that compact, structured domains (KRAS/HRAS, FBXW2) yield the highest druggability scores, while large disordered proteins such as ARID1B are more suited to PROTAC approaches (Supplementary Fig. S10). A composite preclinical score integrating TI (0.40), selectivity (0.30), structure similarity (0.15), and druggability (0.15) ranked ARID1A→ARID1B first (0.817; Fig. 4d; Table 2).

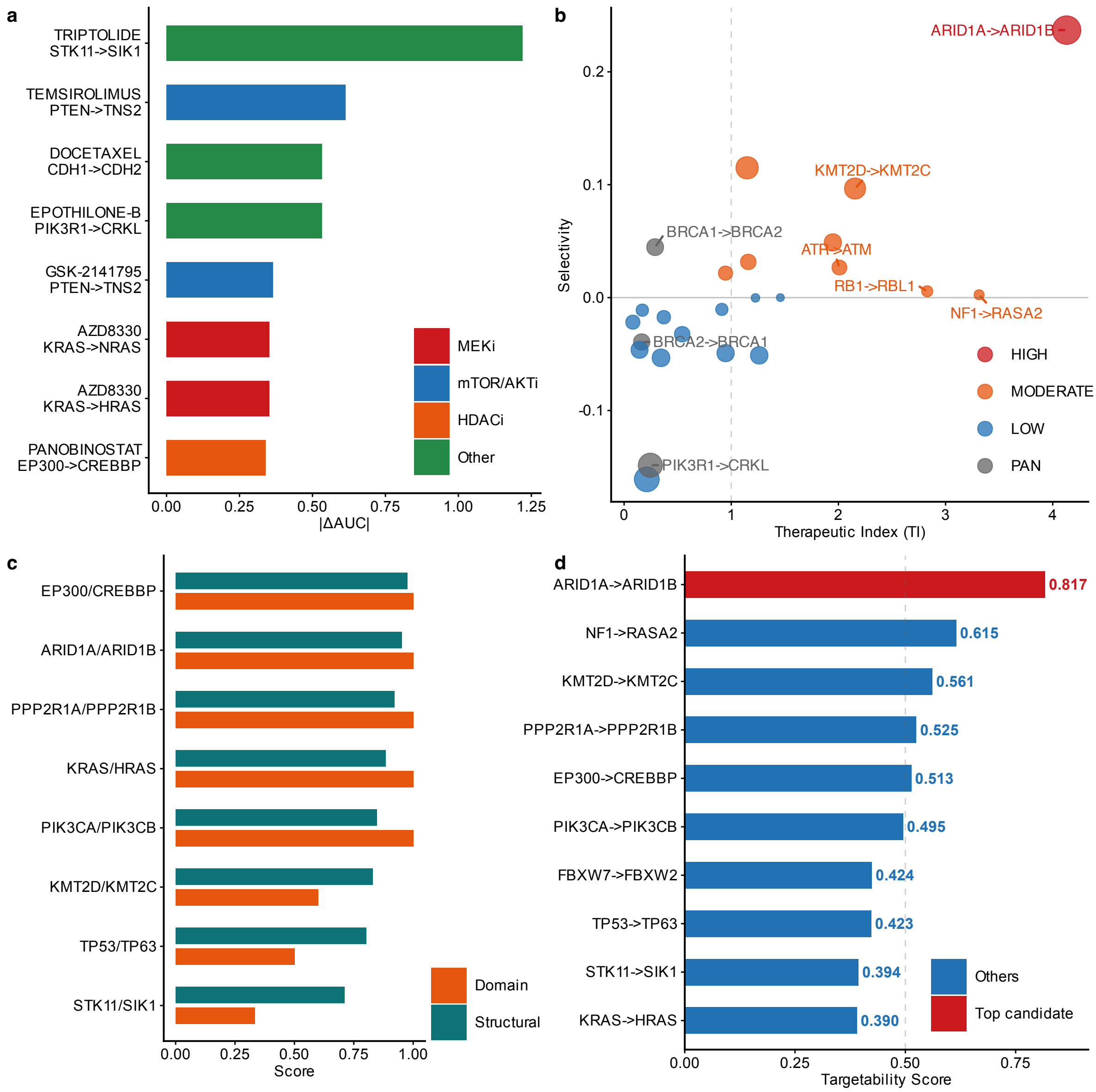

Fig. 4: Pharmacologic context and translational prioritization. **a**, PRISM drug selectivity: —ΔAUC— for top drug-paralog associations. Known drug-target biology is recapitulated: MEK inhibitors (AZD8330, Trametinib) selectively kill KRAS-mutant lines; mTOR/AKT inhibitors (Everolimus, Ipatasertib) kill PTEN-mutant lines; HDAC inhibitors (Panobinostat) kill EP300-mutant lines. BH $q < 0.25$. See Supplementary Fig. S5 for full results. **b**, Therapeutic window classification. Bubble size, mean TI; color, selectivity tier. HIGH_SELECTIVITY requires selectivity > 0.15 and TI > 1.0; NF1→RASA2 has high TI (3.31) but near-zero selectivity (0.003), hence MODERATE. **c**, Structural similarity (top row) and domain conservation (bottom) for 11 key paralog pairs. Filled circles, perfect domain conservation (Jaccard = 1.0). EP300↔CREBBP has the highest structural similarity (0.976). **d**, Composite preclinical prioritization score ranking the top 10 candidates. Scores integrate therapeutic index (0.40), selectivity (0.30), structural similarity (0.15), and druggability (0.15). ARID1A→ARID1B ranks first (0.817).

Table 2: Top paralog-SL candidates ranked by preclinical prioritization score. TI: mean therapeutic index across 5 cancer contexts. Class: HS = HIGH_SELECTIVITY, MOD = MODERATE, LS = LOW_SELECTIVITY, PE = PAN_ESSENTIAL. Struct: structural similarity score. Target.: composite preclinical score. Known SL pairs marked with ⋆. *All candidates are computationally nominated and require experimental validation before clinical translation.*

Driver	**Paralog**	**TI**	**Class**	**Known**	**Struct**	**Target.**
ARID1A	ARID1B	4.13	HS	⋆	0.952	0.817
NF1	RASA2	3.31	MOD		0.658	0.615
KMT2D	KMT2C	2.16	MOD		0.829	0.561
PPP2R1A	PPP2R1B	1.46	LS	⋆	0.921	0.525
EP300	CREBBP	1.15	MOD	⋆	0.976	0.513
PIK3CA	PIK3CB	1.26	LS	⋆	0.847	0.495
FBXW7	FBXW2	1.23	LS	⋆	0.666	0.424
TP53	TP63	0.95	MOD		0.803	0.423
STK11	SIK1	0.91	LS	⋆	0.710	0.394
KRAS	HRAS	0.14	LS		0.885	0.390

Discussion

We set out to test whether a single, interpretable metric — the dependency shift between driver-mutant and wild-type cells — could compete with black-box classifiers for paralog-SL prediction. Across 23 solid tumor types, DD reached AUROC = 0.736 in head-to-head comparison on 77 pairs using one feature, exceeding all four multi-feature classifiers tested (0.563–0.699). The dependency shift appears to carry most of the signal.

Why protein, not RNA? Our seven-cohort CPTAC survey found consistent paralog protein co-variation but an RNA signal indistinguishable from noise (AUROC = 0.348). EP300↔CREBBP was significant in five of seven cohorts; KRAS↔HRAS correlation reached $r = 0.667$ in LUSC. A plausible mechanism is complex stoichiometry: many paralog pairs (EP300/CREBBP, ARID1A/ARID1B) encode subunits of the same multi-subunit complex, and loss of one subunit can stabilize the remaining paralog by reducing competition for complex assembly or proteasomal degradation. This would produce protein-level but not transcript-level co-variation, exactly what we observed. The mutation-conditioned analysis in UCEC is consistent with this model: ARID1B protein was 55% higher in ARID1A-mutant tumors, though the result is suggestive ($p = 0.082$) and needs replication in additional cohorts and pairs. If the stoichiometric model holds, it also provides a rationale for PROTAC-based degradation strategies, which exploit protein abundance rather than transcriptional regulation.

TSG versus oncogene contexts. DD performed better in TSG-driven cancers (mean AUROC = 0.737) than oncogene-driven cancers (0.595). The difference did not reach significance ($p = 0.071$), and with only three oncogene lineages the comparison is underpowered. Biologically, however, the direction makes sense: TSG loss-of-function creates classical synthetic lethality where a paralog compensates for a missing function, whereas oncogene gain-of-function (KRAS, PIK3CA) may produce synthetic dosage lethality — a weaker, more context-dependent dependency shift. For practical purposes, DD is most reliable in TSG-driven contexts; oncogene-driven paralog dependencies likely require pathway-aware models.

Clinical implications. MSI status stratified DD signal strength (MSS ¿ MSI-H in colorectal, $d = 2.72$), and truncating mutations elicited stronger compensation than missense variants for several TSG pairs. For an ARID1B-directed trial, these observations suggest a three-tier enrichment strategy: enroll MSS tumors, require ARID1A truncating mutations, and select patients with low baseline ARID1B expression to maximize the therapeutic window. This is a computational proposal; the enrichment magnitude needs prospective estimation. Our TI framework nominated ARID1A→ARID1B (TI = 4.13) as the leading candidate while flagging BRCA1/2 paralogs as pan-essential. The TI is based entirely on in vitro data and cannot predict organ-specific in vivo toxicity; natural extensions would incorporate tissue expression data from GTEx, mouse knockout phenotypes from IMPC, and germline constraint metrics (pLI/LOEUF) from gnomAD.

Relationship to D'Antonio et al. D'Antonio et al. [16] published the foundational analysis of paralog genetic interactions. Our work extends theirs in three ways. We benchmark DD against eight published deep learning methods in the CV3 setting — a comparison their study did not perform. Our seven-cohort CPTAC proteomic survey reveals protein-level co-variation that their analysis did not examine. And we add clinically actionable stratification (MSI, mutation type) plus a quantitative therapeutic window framework for translational prioritization. Together, the five orthogonal validation layers go beyond a gene-family dependency atlas toward a prioritized, experimentally actionable candidate list.

Data independence. Both DD and the gold-standard labels come from DepMap CRISPR data, which raises a circularity concern. We addressed this with a three-tier strategy. First, removing the two gold-standard pairs with DepMap-era evidence increased AUROC from 0.794 to 0.833; restricting to eight pairs with pre-DepMap experimental evidence raised it to 0.900, confirming the signal predates the dataset. Second, CPTAC proteomics, PRISM drug sensitivity, and TCGA survival provide independent data layers. Third, prospective validation against SL pairs catalogued after the DepMap 26Q1 release is planned.

The ARID1A→ARID1B paradox. Our top-ranked candidate has the largest |DD| (0.267), highest TI (4.13), and highest selectivity (0.237) among all 24 pairs, yet its within-driver BH-corrected q-value is 0.50 — nominally non-significant. This reflects ARID1A's >200 HGNC paralogs creating an extreme multiple-testing burden, not weak signal. The q-value tests whether ARID1A→ARID1B is significantly enriched among all ARID1A paralogs; the multivariate regression ($p = 4.7 \times 10^{-28}$ after controlling for TP53 and CNV) tests whether ARID1B dependency shifts in ARID1A-mutant cells. For clinical prioritization, effect size and selectivity matter more than within-driver FDR ranking; the q-value highlights the gap between statistical significance and biological relevance when sample sizes are small.

Evaluation frameworks. We report two AUROC values. The cancer-type-specific evaluation (AUROC = 0.794; 118 driver×paralog×lineage entries, 11 positives) preserves lineage-specific DD values. The per-pair aggregated evaluation (AUROC = 0.668; 77 unique pairs, 8 positives) averages |DD| across lineages per pair and is more conservative. Bootstrap and permutation analyses use the per-pair framework.

Limitations.

1. *Sample size:* Several rare driver mutations appear in fewer than three DepMap cell lines and could not be analyzed; some lineages lacked sufficient samples for MSI and mutation-type stratification.
2. *Gold-standard heterogeneity:* The evaluation set mixes functional analogs (e.g., BRCA1↔BRCA2,

whose low k-mer Jaccard ≈ 0.23 reflects short 3-mers shared by chance between long proteins rather than true sequence homology) with true sequence paralogs. Stratifying by paralog type in future evaluations would improve clarity.

3. *Circular validation:* DD and the gold standard both derive from DepMap data; the core AUROC measures internal consistency, not independent prediction. CPTAC, PRISM, and TCGA provide partially independent support but do not directly validate SL prediction.

4. *Observational design:* DD captures association, not causation. We controlled for lineage, co-mutations, and CNV through multivariate regression but cannot eliminate residual confounding. CRISPR double-knockout in isogenic models remains the necessary next step.

5. *In vitro therapeutic window:* TI is based on cell line dependencies and cannot predict organ-specific toxicity. Integration with GTEx, IMPC, and gnomAD constraint data is needed.

6. *Structural analysis:* Limited to sequence-level features. AlphaFold-Multimer complex predictions and experimental binding data are needed for structure-guided drug design.

7. *CPTAC coverage:* Not all paralogs were quantified in all cohorts (41–45 of 58 per cohort); mutation-conditioned analysis was performed only for ARID1A→ARID1B in UCEC.

Future directions. (1) CRISPR double-knockout validation of ARID1A→ARID1B in ARID1A-mutant versus wild-type isogenic lines (e.g., TOV-21G, HCT116, MCF7), with viability and colony formation as primary readouts. This study is purely computational; experimental validation through collaboration with functional genomics laboratories is the essential next step. (2) External hold-out validation against SL pairs catalogued in SynLethDB or other databases after the DepMap 26Q1 release. (3) Extending mutation-conditioned proteomic analysis from ARID1A→ARID1B in UCEC to additional paralog pairs and CPTAC cohorts. (4) PROTAC-based targeting of ARID1B, leveraging its high lysine density and intrinsic disorder. (5) Building lineage-specific paralog-SL atlases with MSI-aware filtering for clinical trial biomarker development.

Conclusions

We introduce Delta Dependency, a single-subtraction metric for paralog-based SL prediction, and evaluate it across 23 solid tumor types. DD achieves internal-consistency AUROC of 0.794 for the paralog-SL task within DepMap, comparing favorably with the best published deep learning result in an analogous CV3 setting (SLMGAE [14], 0.790). We note that these evaluations use different test sets; a rigorous head-to-head benchmark would require running all methods on identical paralog-SL data. CPTAC proteomics across seven cohorts reveals protein-level paralog co-variation that is undetectable at the RNA level, and mutation-conditioned analysis in UCEC provides initial evidence for compensatory protein upregulation. MSI status and mutation type offer clinically actionable stratification biomarkers. Quantitative therapeutic window analysis combined with PRISM drug screening and structural druggability assessment nominates ARID1A→ARID1B as the leading computational candidate for preclinical development, pending experimental validation.

We provide a complete open-source toolkit — the `paralogSL` R package and a reproducible Python pipeline — that requires only publicly available DepMap data and no specialized computational infrastructure.

Methods

Data sources and preprocessing

DepMap 26Q1. CRISPRGeneEffect.csv (420 MB; Chronos-normalized gene dependency scores (hereafter "gene effect scores") for 1,208 cell lines × 18,531 genes), OmicsSomaticMutations.csv (554 MB), OmicsExpressionProteinCodingGenesTPMLogp1.csv (483 MB), and Model.csv (681 KB) were downloaded from the DepMap Portal (https://depmap.org/portal/download/, release 26Q1) [32, 33]. Damaging mutations were defined as: Nonsense_Mutation, Frame_Shift_Del, Frame_Shift_Ins, Splice_Site, and Translation_Start_Site [34]. Cell lines without expression or dependency data were excluded, yielding 1,208 cell lines. Cross-study dependency data integration followed established methods [35].

HGNC gene families. The HGNC complete set was downloaded from https://storage.googleapis.com/public-download-files/hgnc/tsv/tsv/. Protein-coding genes with "Approved" status and non-empty `gene_group_id` were retained (11,050 genes). All pairwise gene group combinations yielded 66,595 paralog pairs.

CPTAC proteomics. Ovarian HGSC global proteome data were obtained from the NCI Proteomic Data Commons (PDC study PDC000360, $n = 232$) [29]. Seven additional CPTAC cohorts were retrieved via the cBioPortal API [27, 28]: BRCA ($n = 122$), COAD ($n = 97$), LUAD ($n = 110$), GBM ($n = 99$), PDAC ($n = 140$), UCEC ($n = 95$), and LUSC ($n = 108$). Pearson correlations were computed on $\log_2$-transformed abundance values ($\geq$10 paired samples per test).

PRISM drug sensitivity. The PRISM Repurposing secondary screen dataset (1,482 compounds × 727 cell lines) was downloaded from the DepMap Portal [31]. $\Delta\text{AUC} = \text{mean}(\log_2\text{AUC}_{\text{MUT}}) - \text{mean}(\log_2\text{AUC}_{\text{WT}})$. Selective drugs: BH $q < 0.25$, $\Delta\text{AUC} < 0$, enrichment > 0.1.

TCGA survival data. Pan-cancer overall survival and gene expression data were obtained from the UCSC Xena browser [30]. Breast cancer patients ($n = 1{,}082$) were stratified by median paralog expression. Cox proportional hazards models estimated hazard ratios.

Delta Dependency computation

DD was computed as defined in Equation 1, using Welch's t-test (unequal variances) with Cohen's d as the standardized effect size (pooled SD). Minimum three mutant and three wild-type cell lines per driver. Benjamini-Hochberg correction [36] ($q < 0.20$) was applied within each driver gene: for driver D, all (D, P_i, c_j) combinations were corrected jointly. This is a discovery-stage pipeline; global FDR correction across all 66,595 pairs × 40 drivers × 23 cancer types was not performed because the goal was to maximize sensitivity for candidate identification. Candidates should be treated as hypothesis-generating. At a stricter $q < 0.10$, PIK3CA→PIK3CB remained significant ($q = 0.086$ in Endometrial), whereas ARID1A→ARID1B did not survive within-driver correction ($q = 0.50$), not because of weak signal ($|\text{DD}| = 0.30$) but because ARID1A has >200 HGNC paralogs.

Benchmark comparison

Gold-standard positives and matched negatives. Twelve paralog-SL pairs were curated from published experimental studies (genetic screens, viability assays, biochemical validation) independently of any DepMap-derived results (Supplementary Table S3). Of these, 10 are true sequence paralogs (shared gene family, conserved domains), 1 is a partial homolog (STK11→SIK1), and 1 is a functional analog (BRCA1↔BRCA2; included as a mechanistic comparator). Negative pairs comprised all remaining HGNC-defined paralog pairs involving the same 40 driver genes that did not appear in the positive set. This yields an imbalanced evaluation set (12 positives vs. ∼194 negatives per lineage), which is representative of the real-world class imbalance but may inflate AUROC; we therefore also report AUPRC where applicable. Directional pairs (e.g., BRCA1→BRCA2 and BRCA2→BRCA1) are scored separately when both driver genes are in the 40-gene panel.

Contextual comparison with published methods.

DD was compared against eight published SL prediction methods evaluated in the CV3 setting by Feng et al. [13]. Published AUROCs are from the CV3 evaluation reported in Supplementary Data 1 of that study (NSMRand negative sampling, 1:1 positive:negative ratio, complete dataset); DD was evaluated on the paralog-SL prediction task using the 12 gold-standard paralog-SL pairs as the positive set. AUROC was computed using scikit-learn 1.3.0 [37]. Bootstrap 95% CI: 1,000 iterations, resampling paralog pairs with replacement. For the DD + ID≥30% subset (10 pairs), bootstrap CI was computed separately to account for the reduced sample size. Negative controls: 100 permutations of shuffled known-SL labels.

Sequence identity computation

K-mer-based Jaccard similarity ($k = 3$) was used as a computationally efficient proxy for global protein sequence identity. This metric was validated against Needleman-Wunsch global alignment identity in a subset of 50 protein pairs (13 known paralog pairs and 37 non-paralog controls; Pearson $r = 0.88$, $p < 10^{-16}$; Supplementary Fig. S9). Within-group correlations were $r = 0.91$ (paralogs, $n = 13$) and $r = 0.41$ (non-paralogs, $n = 37$), confirming that the high overall r is not solely driven by between-group variance. The ≥30% threshold should therefore be interpreted as an approximate filter for enriching high-identity pairs rather than a precise sequence identity cutoff.

Potential confounders and causal interpretation

DD quantifies an observational association between driver mutation status and paralog dependency, not a causal effect. Potential confounders include: (1) lineage differences between mutant and wild-type cell lines within a cancer type; (2) co-occurring mutations that may independently affect paralog dependency; (3) copy number alterations at paralog loci. We mitigated these through: within-lineage analysis (DD computed separately per cancer type), Welch's t-test (robust to unequal variances), and CNV independence verification (Supplementary Fig. S4, all $R^2 < 0.10$). Benjamini-Hochberg correction was applied within each driver gene to control false discovery rate. Nevertheless, experimental validation (e.g., CRISPR double-knockout in isogenic cell lines) remains essential to establish causality.

CPTAC proteomic co-variation

Protein abundance data for seven CPTAC cohorts (BRCA, COAD, LUAD, GBM, PDAC, UCEC, LUSC; $n = 771$ total) were retrieved via the cBioPortal API [27, 28]. For each paralog pair, Pearson correlation of protein $\log_2$-abundance was computed across all samples within each cohort. Significance was assessed using BH correction within each pair across 7 cohorts. Mutation-conditioned proteomic analysis was performed for the ARID1A→ARID1B pair in the UCEC cohort: ARID1A mutation status was retrieved from the cBioPortal mutations API (`ucec_cptac_2020_mutations`), and ARID1B protein abundance was compared between ARID1A-mutant ($n = 37$) and wild-type ($n = 58$) samples using the Wilcoxon rank-sum test. Extension to additional pairs and cohorts is a priority future direction.

Pan-cancer lineage definitions

Twenty-seven solid tumor types were defined by matching OncotreePrimaryDisease annotations to curated keyword lists; the 23 types with ≥ 6 cell lines with available data were analyzed (Supplementary Table S1). Lineage short labels follow DepMap/Oncotree conventions and mix organ-based and histology-based names as required by each cancer type's classification (e.g., NSCLC and SCLC are distinct histological subtypes of lung cancer); full names are given in Supplementary Table S1. Forty pan-cancer driver genes were selected from TCGA pan-cancer analyses [23] and the COSMIC Cancer Gene Census [24], filtered to genes with ≥ 3 mutant cell lines in at least one lineage (Supplementary Table S5).

MSI stratification

MSI-H was approximated by the presence of damaging mutations in MMR genes (MLH1, MSH2, MSH6, PMS2) or POLE as a proxy for microsatellite instability status; this mutation-based proxy may not perfectly correspond to clinical MSI testing (PCR or IHC). DD computed separately within each subgroup for colorectal and endometrial cancers.

Mutation type classification

Driver mutations classified as truncating (frameshift, nonsense, splice-acceptor, splice-donor, start-loss, stop-loss) or missense (missense_variant, protein_altering_variant, excluding truncating) using VEP VariantInfo annotations. Per-cell-line classification used the most severe mutation type.

Therapeutic window analysis

TI defined as in Equation 2. The denominator uses $\max(|\mu_{\text{CERES}}|, f_{\text{pan-essential}}, 0.01)$ to capture the most conservative estimate of baseline paralog essentiality: $|\mu_{\text{CERES}}|$ reflects mean dependency magnitude (dimensionless, normalized by Chronos), $f_{\text{pan-essential}}$ reflects the fraction of lines where the paralog is essential (dimensionless, range 0–1), and the 0.01 floor prevents division by near-zero values for non-essential paralogs. All three quantities are dimensionless and on comparable scales (0–1); the max operator selects whichever indicates the greatest baseline toxicity risk. Paralogs classified: PAN_ESSENTIAL ($f_{\text{pan-essential}} > 0.5$), HIGH_SELECTIVITY (selectivity > 0.15 and TI > 1.0), MODERATE, or LOW_SELECTIVITY. Analysis across 5 cancer contexts (Ovarian, Endometrial, Breast, Colorectal, PanCancer).

Structural druggability

Sequence-based features (length, MW, pI, GRAVY hydrophobicity, disorder propensity, lysine/cysteine density) from UniProt canonical sequences. Structural similarity: $1 - \overline{|\Delta\text{feature}|}$. Domain architectures from UniProt annotations. PROTAC suitability: lysine_density $\times$ 2.0 + disorder_propensity $-$ cysteine_density $\times$ 0.5. Composite targetability: therapeutic index (0.40), selectivity (0.30), structural similarity (0.15), druggability (0.15).

Statistical analyses

AUROC via scikit-learn 1.3.0 [37]. Bootstrap CI: 2.5th and 97.5th percentiles from 1,000 iterations. Negative controls: 10,000 label permutations. Analyses in Python 3.10 with numpy 1.24 [38], scipy 1.11 [39], pandas 2.0, and statsmodels 0.14. The companion R package `paralogSL` provides DD computation, PCS analysis, benchmark comparison, and CPTAC correlation visualization under the MIT license.

Power analysis and test set stratification

Gold-standard pairs were stratified into two evaluation sets: a *Primary set* of 10 true sequence paralogs (shared gene family and conserved domains) and a *Secondary set* of 2 functional analogs (BRCA1$\leftrightarrow$BRCA2, STK11$\rightarrow$SIK1) included as mechanistic comparators. Main AUROC results report the Primary set; the Full set (12 pairs) is reported for completeness. With 8–10 positive pairs, the permutation test has limited power: at $\alpha = 0.05$, an estimated $\geq$25 validated positive pairs would be needed to achieve 80% power for detecting AUROC = 0.85 against a null of 0.5. To compensate for this limitation, we complement binary classification metrics with effect-size ranking (—DD—, TI, selectivity) and multivariate regression significance (p-values for individual pairs).

Supplementary information

Supplementary Figures S1–S10 and Supplementary Tables S1–S6 are available as a separate PDF file. Supplementary Fig. S1: Panoramic baseline across 23 solid tumor types (cell line counts and tested pairs, colored by AUROC tier). Supplementary Fig. S2: Cross-cancer DD AUROC detailed values. Supplementary Fig. S3: CPTAC per-cohort scatter plots. Supplementary Fig. S4: CNV independence analysis. Supplementary Fig. S5: PRISM drug selectivity full results. Supplementary Fig. S6: Full mutation-type stratification. Supplementary Fig. S7: Therapeutic window full classification. Supplementary Fig. S8: Full bootstrap distribution and negative control. Supplementary Fig. S9: k-mer Jaccard vs. Needleman-Wunsch alignment identity validation (n = 50 pairs, Pearson r shown). Supplementary Fig. S10: Structure-based druggability assessment of 15 proteins using ESMFold-predicted structures. Supplementary Table S1: Cell line counts per cancer type. Supplementary Table S2: Complete paralog-SL pair analysis results (118 driver$\times$paralog$\times$cancer-type associations passing the minimum sample size filter of $\geq$3 mutant and $\geq$3 wild-type cell lines). Supplementary Table S3: Gold-standard paralog-SL pair evidence. Supplementary Table S4: Pan-cancer summary statistics. Supplementary Table S5: Complete 40-gene driver panel with cancer types, selection source, and functional category. Supplementary Table S6: All 24 paralog pairs ranked by preclinical prioritization score with component values.

Declarations

Ethics approval and consent to participate

Not applicable. This study used exclusively publicly available, de-identified data.

Data availability

All data are publicly available: DepMap 26Q1 from https://depmap.org/portal/download/; CPTAC proteomics from https://cbioportal.org and https://pdc.cancer.gov; TCGA PanCan from https://xenabrowser.net; PRISM from https://depmap.org/portal/prism/. HGNC gene families from https://www.genenames.org/. Processed datasets with cell line-level DD values and all supplementary tables are archived at Zenodo (DOI: to be assigned upon acceptance) and the GitHub repository below, enabling direct integration with clinical genomic pipelines.

Code availability

All analysis code is available under the MIT license at https://github.com/tjogzt/paralogSL (DOI: 10.5281/zenodo.21502113). The complete analysis pipeline (DOI: 10.5281/zenodo.21502030) is available at https://github.com/tjogzt/paralog-sl-predictor. The paralogSL R package (v1.0.1, 19 exported functions) provides DD computation, benchmark comparison, therapeutic window analysis, CPTAC visualization, and clinical decision support (`predict_trial_response`, `visualize_dependency_shi`
Random seeds (`set.seed(42)` in R, `random_state=42` in Python) are fixed throughout.
Workflow: DepMap gene-effect matrix → `build_mutation_matrix()` → `compute_dd()` per driver×paralog×lineage → `compute_auroc()` against gold standard → `compute_therapeutic_window()` for translational ranking → visualization via `plot_pancancer_summary()` and `plot_therapeutic_window()`.
Quick start (3 lines):

```
install.packages("paralogSL")  # or devtools::install_github("taozhu/paralogSL")
library(paralogSL)
result <- run_paralog_analysis(dep_matrix, mut_matrix, driver="ARID1A", paralog="ARID1B")
```

The package includes built-in datasets (`known_sl_pairs`, `gyn_drivers`, `solid_tumor_summary`), a vignette with worked examples, and 19 exported functions. A frozen release with all processed data tables is archived at Zenodo (https://doi.org/10.5281/zenodo.21502030). The complete Python analysis pipeline (`main.py`, `pcs.py`, `pancancer.py`) is included in the same repository for full reproducibility. In addition, three audit scripts (`compute_headline_metrics.py`, `ml_benchmark.py`, `regression_controls.py`) recompute every quantitative claim in this manuscript directly from raw DepMap data — headline metrics, classifier benchmarks, and covariate-adjusted regressions, respectively — each ending with an automated claim-by-claim comparison against the reported values; a single entry point (`verify_all.sh`) runs all three audits plus the test suite in under one minute.

Competing interests

The authors declare no competing interests.

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Authors' contributions

Q.Q.M. and T.Z. conceived the study. T.Z. developed the methodology, performed all computational analyses, and developed the R package. Q.Q.M. curated clinical datasets and provided gynecological oncology domain expertise. Q.Q.M. and T.Z. wrote the manuscript.

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